

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456923.8543 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.071 +/- 0.019
Transit depth (Rp/Rs)^2	0.5 +/- 0.27 [%]
Orbital Inclination (inc)	89.24 +/- 1.4
Airmass coefficient 1 (a1)	0.9897 +/- 0.0069
Airmass coefficient 2 (a2)	0.0094 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	#2 - [42, 352]
Optimal Aperture	3.48
Optimal Annulus	4.56
Transit Duration (day)	0.1483 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.071 ± 0.019 vs 0.1036 (deviation 1.72σ)
Duration fit vs published	0.1483 d vs 0.1567 d (deviation 2.39σ)
Mid-time O–C	4.8 min
Depth SNR	3.7
Residual scatter	0.7 %

Light curve

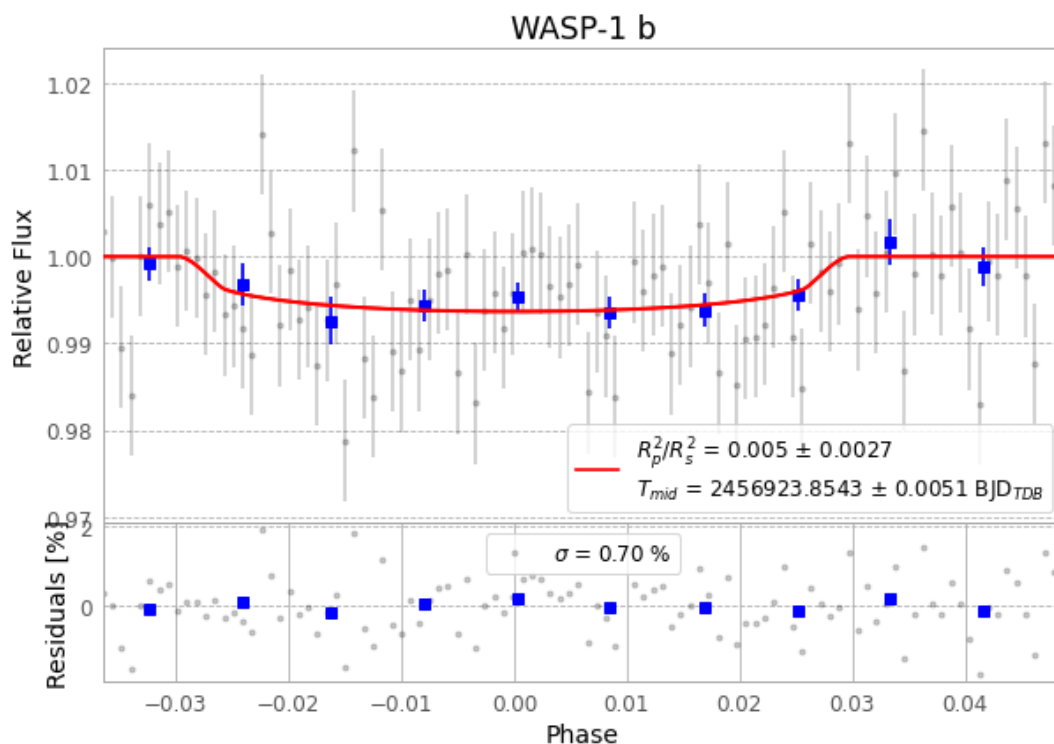


Figure 1 — FinalLightCurve_WASP-1 b_2014-09-22.png (SHA-256 390398cdb0db45ff...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [330, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a60082f08f2a21a0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

104 FITS frames (68 MB), WASP-1140923060912.FITS → WASP-1140923111813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-140923.log (00c6db11871b...), FinalLightCurve_WASP-1 b_2014-09-22.png (390398cdb0db...), FinalLightCurve_WASP-1 b_2014-09-22.csv (def8f749976c...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 15:10Z.